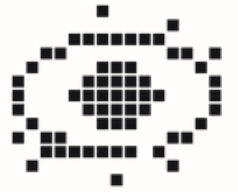


guardatum



Hidden cost *of free browsing*



Abstract

Privacy policies are the legal instrument through which consent is obtained. We tested whether they describe reality. Across the 100 most-visited websites in Europe, we measured what each site does on arrival (before any consent is given) and separately scored its written privacy policy against the data-protection principles of GDPR Article 5. The two are unrelated.

Policy quality showed no meaningful correlation with third-party tracking ($r = 0.05$), with cookies set before consent ($r = 0.04$), or with whether a site offers any means of refusal ($r = -0.14$). Meanwhile 94% of measured sites set cookies before a visitor interacts with anything, 71% provide no first-layer way to reject, and reading all 77 unique policies in the sample once would take 35 hours.

A well-written privacy policy is not evidence of privacy-respecting behaviour. It is evidence of a well-written privacy policy.

Why this study exists

The General Data Protection Regulation (GDPR) rests on a premise: that people are informed, and that on the basis of being informed, they consent. The privacy policy is where the informing is supposed to happen. The cookie or privacy banner is where the consenting is supposed to happen. Both are widely assumed to be fictions. What has been missing is measurement — specifically, measurement that compares the **document** against the **behaviour**.

We treat the privacy policy as a claim, and the site's observable conduct as evidence about that claim. We then ask a question that has not, to our knowledge, been asked at this scale for the European context: **does a well written privacy tell us what the company does with our data?**



Related work

Four elements of prior research are directly relevant to this study: the automated analysis of privacy-policy text, the measurement of consent interfaces, the measurement of tracking behaviour, and the emerging use of large language models to read policies at scale. Each of these fields in research has advanced considerably. However, what none has done, to our knowledge, is place the document and the behaviour side by side, on the same sites, and ask whether one can explain the other.

McDonald and Cranor's foundational estimate put the time an average user would need to read the policies of the sites they visit in a year at roughly **201 hours**, with a national opportunity cost they valued in the hundreds of billions of dollars, this study was conducted in an era when a typical user visited far fewer sites than today [McDonald & Cranor 2008]. Subsequent work established that the problem has not eased. Analysing a corpus of thousands of policies from inside and outside the EU, Linden et al. found that policies grew markedly longer after the GDPR came into force, even as their coverage of GDPR-relevant topics improved [Linden et al. 2020]. A parallel line of work built the annotated corpora that made systematic policy analysis possible in the first place, most notably OPP-115, 115 policies segmented and labelled by legal experts [Wilson et al. 2016], and more recently GDPR-oriented taxonomies such as GoPPC-150 [Tang et al. 2024]. Our study takes the length and readability of policies as an input: we treat the policy as a claim to be tested against conduct, and our own reading-time figures update McDonald and Cranor's for a contemporary European sample.

The cookie banner is the second site where the law's promise is meant to be kept. Degeling et al.'s study of the GDPR's arrival scanned hundreds of the most-visited sites across EU member states and documented both the proliferation of consent notices and the persistent absence of usable mechanisms to refuse rather than accept [Degeling et al. 2019]. A following substantial literature has examined consent-management platforms, the prevalence of dark patterns in banner design, and compliance with the IAB's Transparency and Consent Framework. Our study can be situated in this tradition, and our findings – that a large majority of sites set cookies before any interaction, and that most offer no first-layer means of rejection – are consistent with it. Our contribution here is the pairing of a measuring technique with the policy score.



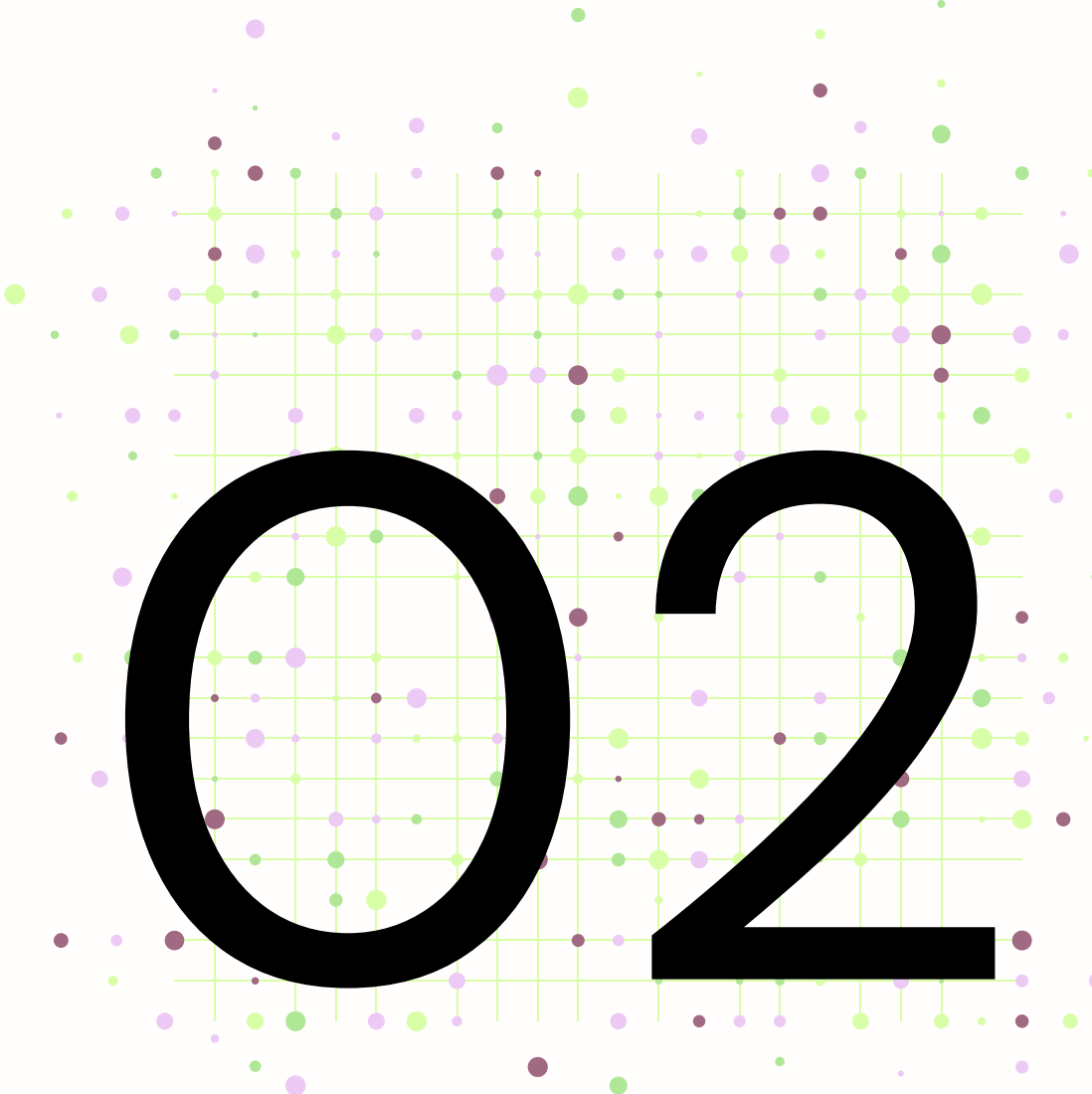
Related work

The census of third-party tracking is a mature field. Englehardt and Narayanan's million-site measurement remains the reference point: using their open-source OpenWPM tool, they crawled the top million sites and characterised stateful and stateless tracking, cookie syncing, and the striking concentration of the tracking ecosystem in a handful of companies — Google-owned domains foremost among them [Englehardt & Narayanan 2016]. Our §4.5 reproduces that finding on a European top-100 sample a decade later: Google/Alphabet accounts for roughly a quarter of all third-party observations, and a small number of companies recur across the corpus. Our tracker counts are deliberately a floor (homepage-only, pre-consent), and we use them not as an end in themselves but as one axis of the correlation that is our subject.

Most recently, a fast-growing body of work has turned large language models on privacy policies. Tang et al.'s PolicyGPT reported that GPT-4 classified policy segments against the OPP-115 and an Article-13-derived taxonomy with accuracy well above earlier machine-learning baselines [Tang et al. 2023]. Chen et al. conducted a systematic evaluation of LLM-based policy classifiers across four corpora, combining prompt engineering with fine-tuning to exceed prior state-of-the-art results, and additionally assessed the explainability of the models' judgements [Chen et al. 2025]. Mori et al. took a different angle, using LLMs to assess policy understandability: across carefully obfuscated policies and a 449-participant user study, they found LLMs comprehended policies substantially better than users (85.2% versus 63.0% correct), while noting the gaps (technical terms and dispersed information) where both struggled [Mori et al. 2025]. This establishes that LLM-assisted reading of privacy policies is a validated technique, which is the ground on which our own LLM-assisted scoring (§2.1, §2.4) stands; and it supplies the appropriate cautions such as central-tendency effects and sensitivity to prompt and phrasing.

Each of these four pillars measure one thing well, whether that's the policy's text, the consent interface, the tracking behaviour or a model's reading of the document. The question that falls between them is whether the quality of what a company writes bears any relationship to what its site actually does. The tools now exist to ask it: LLM-assisted scoring makes policy-quality assessment tractable at scale (from the fourth strand), and automated crawling makes behavioural measurement routine (from the second and third).

METHOD



2026



2.1

Two independent assessments

Each site was assessed twice. It was first measured with an instrumented Chromium browser which loaded each homepage from a clean profile and recorded: every distinct third-party domain contacted; cookies set before any consent interaction; which consent management platform was present; whether the IAB Transparency & Consent Framework (`__tcfapi`) or Global Privacy Platform (`__gpp`) interfaces were exposed; whether a first-layer "reject all" control existed; and which permission APIs the page invoked. The linked privacy policy was fetched separately, its text extracted, word-counted, and hashed.

The crawler is open source and the dataset is published to enable reproduction of the results.

Furthermore, the crawled results were reviewed and each site's privacy policy was scored 0–10 against each principle using a structured rubric applied with large-language-model assistance under author review.

The full rubric with the criteria and keyword sets for each principle is reproduced in Appendix B.

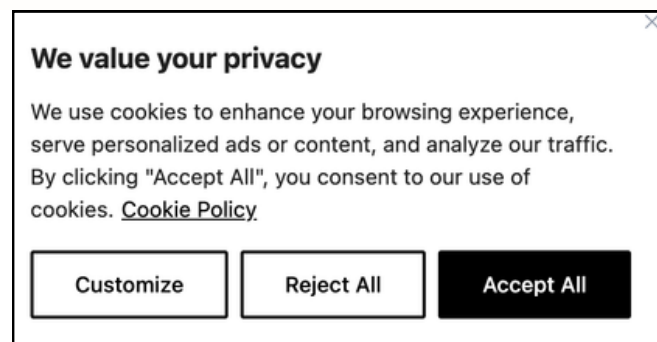


2.2

The pre-consent state

We measure what a site does **on arrival** — before any banner is dismissed, accepted, or rejected. This is deliberate. It is the state every visitor meets, and the state in which the law is most explicit: under Article 5(3) of the ePrivacy Directive, storing or accessing non-essential information on a user's device requires prior consent.

It also means our tracking figures are **floors, not ceilings**. After consent, and on interior pages, sites load substantially more.



Example of a cookie and privacy banner.

2.3

Provenance

Crawl window	19-21 July, 2026
Browser	Chromium 149.0.7827.55
Automation	Playwright 1.61.1
Code commit	`63daf9d`
Sites attempted	100



2.4

Limitations

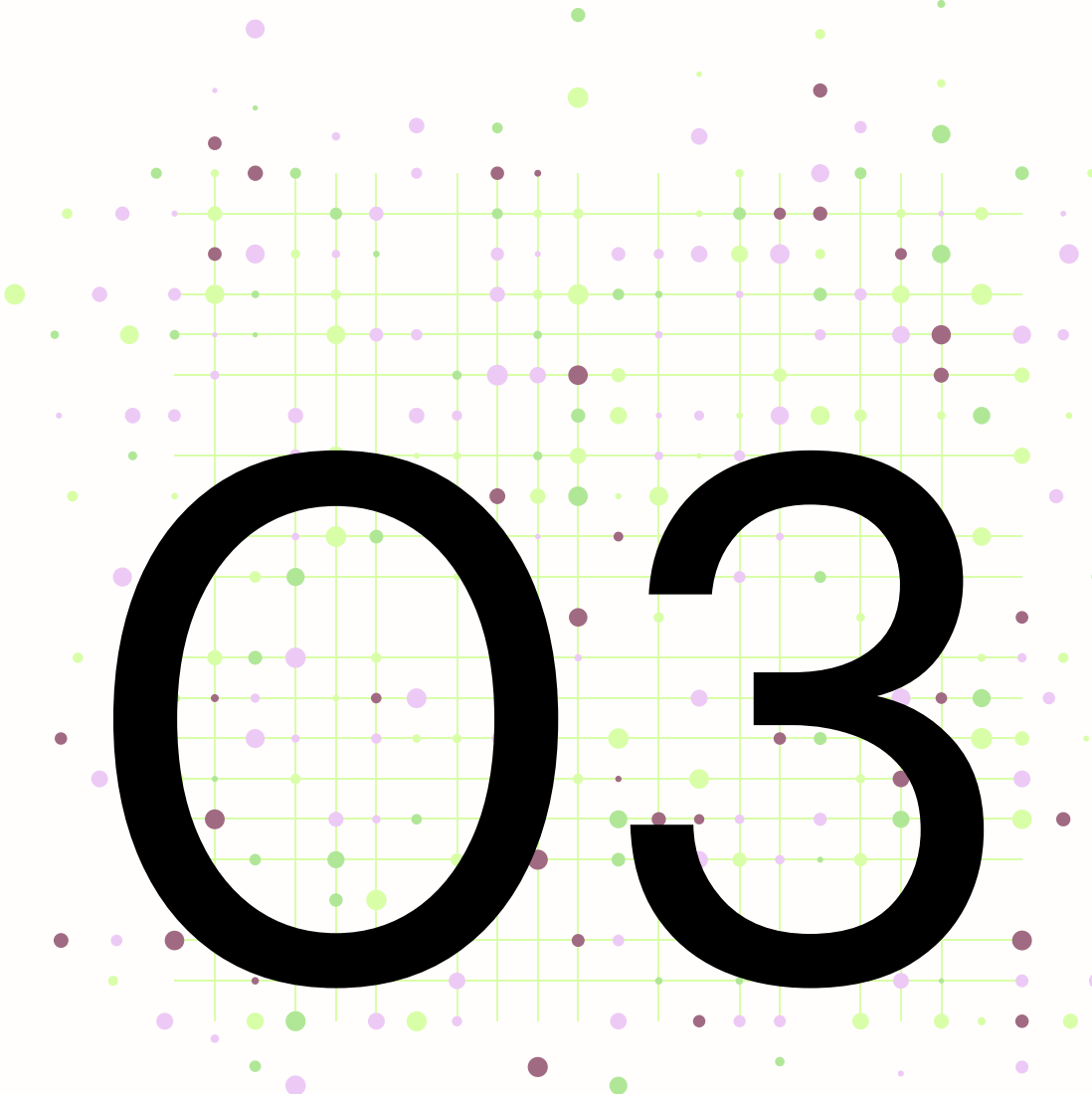
On the measured figures:

- **Homepage only.** Interior pages typically load more third parties.
- **Logged out.** Privacy settings of authenticated sessions often differ.
- **Pre-consent.** A floor, by design.
- **One observation per site.** No longitudinal component; sites change.
- Six sites could not be measured at all.

On the reviewed scores:

- **Restricted range.** Scores span 5.86–9.07 on a nominal 0–10 scale, with a standard deviation of 0.72. Sites differing by a point or less are not meaningfully distinguished by this instrument.
- **Instrument effects.** Language-model-assisted scoring is known to cluster toward central values and away from extremes. Some of the restricted range above is likely an artefact of the instrument rather than a property of the corpus. Restricted range attenuates correlation coefficients.
- **No inter-rater reliability established.** Each policy was scored once. We have not measured the consistency of repeated scoring, and future work should.
- **Selection floor.** The corpus consists of sites that have a published privacy policy at all, which sets a lower bound on how badly any of them can score.

RESULTS



03



Of 100 sites attempted:

- 94 homepages successfully measured
- 84 privacy policies retrieved (77 unique documents)
- 17 sites produced at least one error

3.1

Six sites block automated inspection

H&M, Zara, Sephora, ASOS, Douglas.de, and EasyJet could not be measured. In each case the homepage returned a bot-detection or web-application-firewall interstitial rather than the site; in five cases the policy URL returned HTTP 403.

These sites are excluded from every measured statistic. They are **not** counted as "zero trackers". The implication is that **a subset of Europe's most-visited retailers actively prevents automated inspection of their privacy practices.**

There are legitimate reasons to deploy such defences — scraping, fraud, credential stuffing. The effect, regardless of intent, is that independent verification of their data-protection conduct is not possible by these means.

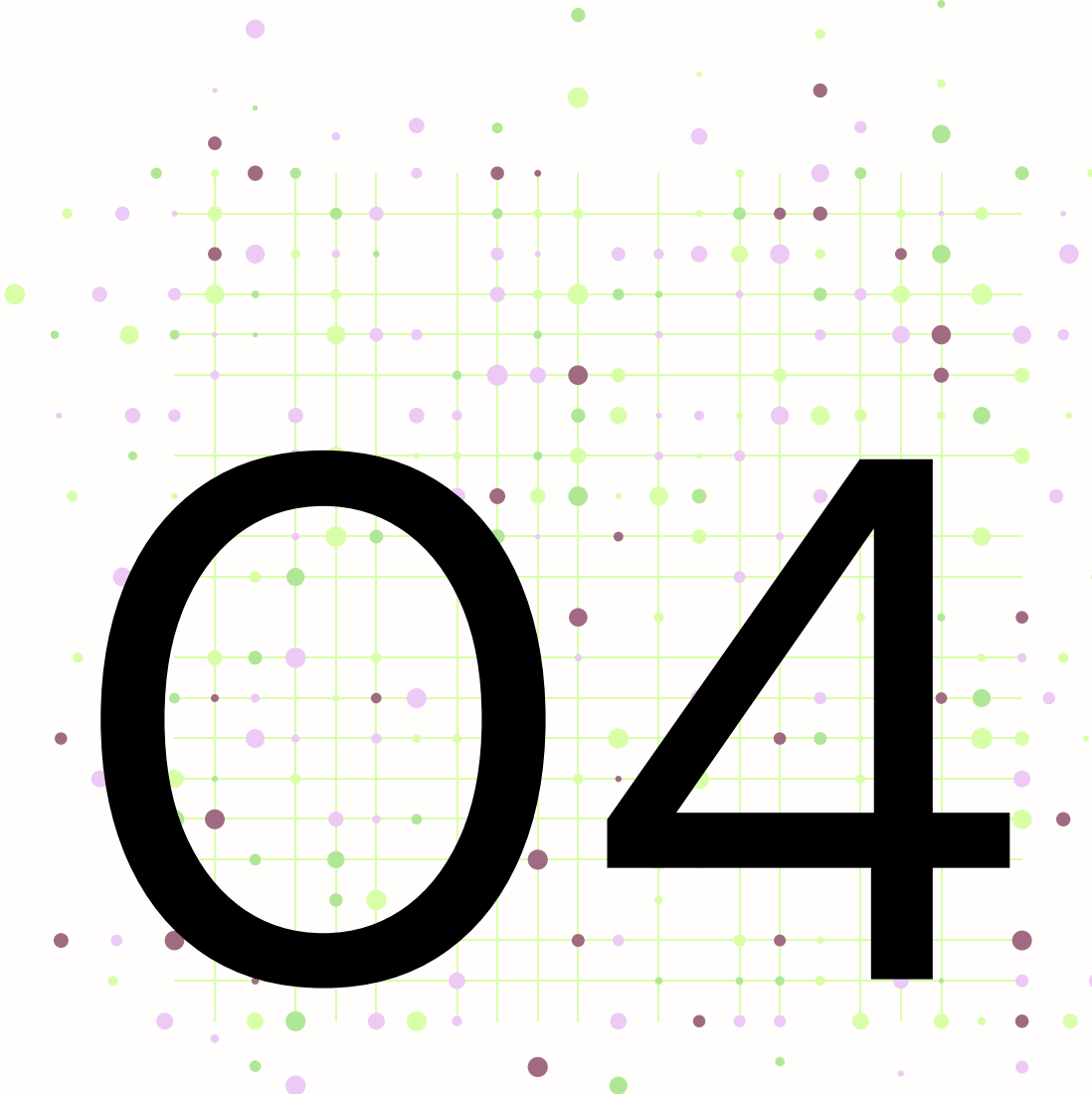
3.2

Policies that could not be read

Eleven further policies could not be extracted: consent walls, JavaScript-gated documents, and in two cases (N26, Zara) a policy served only as a PDF download.

A privacy policy that a standard automated browser cannot read is one that assistive technology may also struggle with. Accessibility of the disclosure is a precondition of the disclosure meaning anything.

FINDINGS



04



4.1

The disclosure is unreadable

Measured (n = 84 policies, 77 unique)

Words	
Shortest	31
25th percentile	3046
Median	5770
75th percentile	9250
Longest	17726

At roughly 230 words per minute, the **median policy takes about 25 minutes to read**; the mean, 27.5. A quarter exceed 40 minutes. The longest exceeds 75. To read every unique policy in this sample once would take 35.3 hours (almost a working week). Several companies serve one document across multiple products (one Google policy governs Search, YouTube, Workspace and Pay; one Amazon policy covers Amazon and Prime Video), and each is counted once here.

That deduplication is itself informative: **a single privacy policy governs four of the most-visited services in Europe.**

Policy word-count

Informed consent, at this length, is not a demanding standard that users fail to meet. It is a standard nobody meets, including the people who wrote the documents.

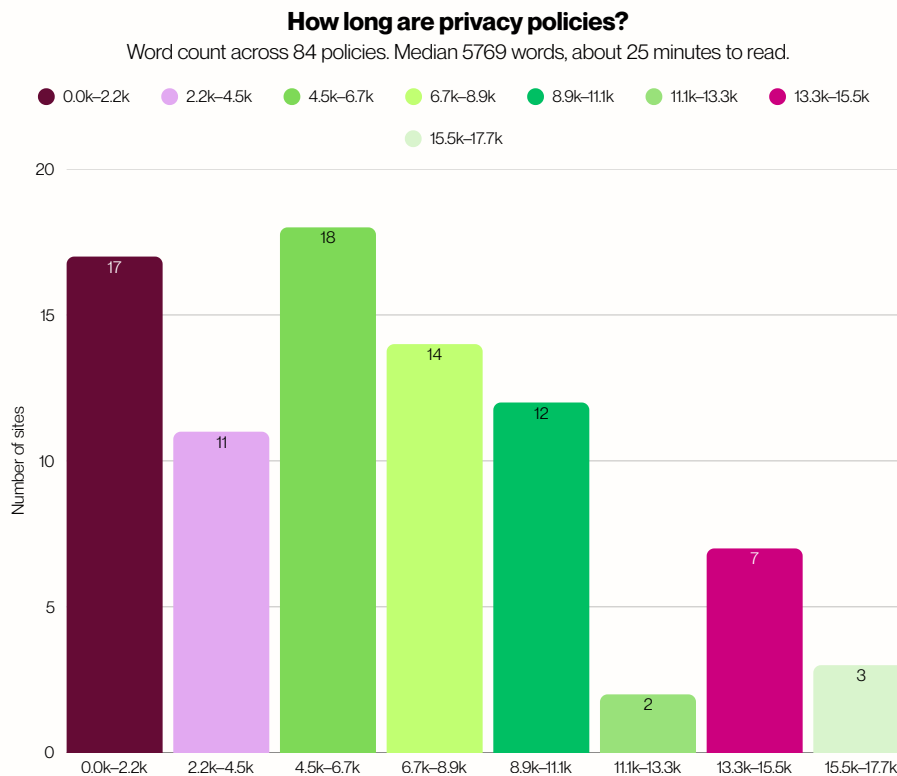


Figure 1: Distribution of privacy-policy word counts across 84 policies. The median policy runs 5,770 words (25 minutes of reading) and the distribution's long right tail reaches beyond 17,000 words.

4.2

Consent is taken before it is given

Measured (n = 94 sites)

94% of measured sites set at least one cookie before the visitor interacted with anything.

Cookies set pre-consent	
25th percentile	3
Median	6
75th percentile	11
Maximum	36

Article 5(3) of the ePrivacy Directive requires prior consent for storing non-essential information on a user's device. Some share of these cookies will be strictly necessary and therefore exempt. But a median of six and a maximum of thirty-six, before a visitor has clicked anything is difficult to reconcile with a regime built on **prior** consent.



4.3

There is frequently no way to refuse

71% of measured sites (67 of 94) **offered no first-layer "reject all" control.** Accepting was one click but refusing required a secondary interface, or was not offered at all.

European Data Protection Board guidance on deceptive design patterns, and multiple national supervisory authority decisions, hold that consent is not freely given where refusal is materially more burdensome than acceptance.

4.4

The standards barely exist in practice

Signal	Sites	Share
Any recognised CMP	38	40%
IAB TCF ('__tcfapi')	13	14%
Global Privacy Platform ('__gpp')	2	2%

Only 40% of sites used a recognised consent management platform. **Then, only 14% exposed the IAB Transparency & Consent Framework** – the advertising industry's own interoperability standard – and 2% implemented GPP.

The majority of consent interfaces on Europe's most-visited sites are bespoke. They cannot be audited programmatically, answered programmatically, or compared across sites. **The absence of a machine-readable consent layer is what makes consent unenforceable at scale.**

Where a CMP was present, the market is concentrated: **OneTrust alone accounts for 28% of all measured sites** – more than every other platform combined.



4.5

The web reports to very few companies

Measured · n = 94 sites, 594 site-domain observations.

Third party	Sites	Share
googletagmanager.com	38	40%
gstatic.com	26	28%
google.com	25	27%
cookielaw.org (OneTrust)	23	24%
onetrust.com	20	21%
googlesyndication.com	15	16%
google-analytics.com	14	15%
googleapis.com	12	13%
cloudfront.net (Amazon)	11	12%
doubleclick.net (Google)	10	11%

Grouped by parent company:

Company	Observations	Share of all
Google/Alphabet	146	24.6%
Amazon	21	3.5%
Meta	5	0.8%
Microsoft	5	0.8%

Roughly one in four of every third-party connection we observed went to Google. Google Tag Manager alone appeared on 40% of sites — the single most widespread third party in the corpus, and notable because it is a **conduit**: a mechanism through which further trackers are loaded without appearing in a site's own source.



4.6

The consent layer is itself a third party

The fourth and fifth most widespread third parties here are not advertising networks. They are **OneTrust** – the consent management platform – served from ``cookie law.org`` and ``onetrust.com``, together present on roughly a quarter of sites.

The apparatus through which European users are invited to **control** tracking is, in practice, another third-party connection to a US vendor, established before any consent is given.

4.7

The extremes

Fewest third parties		Most third parties	
Signal.org	0	Amazon (EU)	45
Telegram.org	0	Amazon Prime Video	42
Apple TV+	0	Euronews.com	29
Apple Music	0	SoundCloud.com	19
BlaBlaCar.com	0	Wire.com	18

Several major services load **zero** third parties, showing that the prevailing pattern is a choice and not a technical necessity.



5.1

The central finding: policy quality does not predict conduct

Reviewed × measured · n = 90 sites.

If privacy policies described practice, we would expect sites scoring well on transparency, minimisation and accountability to load fewer third parties, set fewer cookies before consent, and be more likely to offer a means of refusal.

Relationship	Pearson's R	n
Policy score × third-party count	0.05	90
Policy score × cookies before consent	0.04	90
Policy score × has first-layer reject-all	-0.14	90

All three are statistically indistinguishable from noise. The only coefficient with any sign at all points the **wrong** way: sites with better-written policies were marginally **less** likely to offer a working reject-all control.

5.2

Robustness

During the analysis we identified and corrected several data-integrity errors: six sites whose failed page loads had been recorded as genuine zeros, three block pages misread as policy text, and a denominator error affecting several rates.

Correcting all of them moved the coefficients by hundredths – from 0.03, 0.06, -0.14 to 0.05, 0.04, -0.14.



5.3

What evidence this sets for privacy practices

This report does not show that policies are dishonest. A policy may accurately describe extensive data collection. A high score reflects the quality of the **disclosure**, not the restraint of the **practice**. A site can score well precisely by being candid about how much it takes.

It does not show causation. As set out in §2.4, the reviewed scores occupy a narrow range (sd 0.72), partly because language-model-assisted scoring clusters toward the centre. Restricted range attenuates correlation. A more discriminating instrument might detect a weak relationship where we detect none. What restricted range cannot easily explain is the **consistency** of the null across three independent behaviours, or the negative sign on reject-all.

This report does show that a reader cannot use policy quality as a proxy for conduct. The document and the behaviour carry independent information.

The compliance apparatus European users encounter, namely the policy, the banner, the disclosure is not connected to the conduct it purports to govern.



5.4

Websites perform well on what is visible

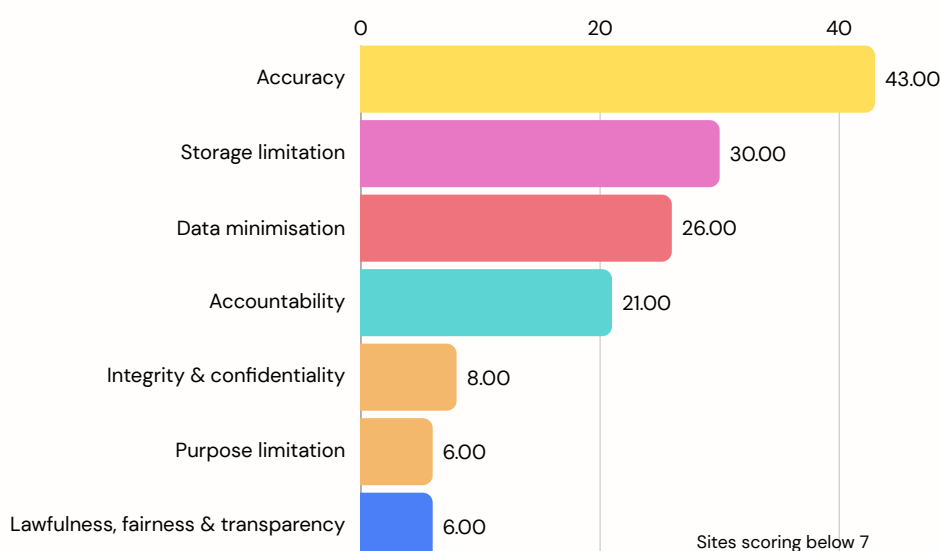
Reviewed · n = 89

GDPR Art. 5 (1) principle	Mean	Median	Scoring below 7
Lawfulness, fairness, transparency	8.12	8	6
Purpose limitation	7.92	8	6
Integrity and confidentiality	7.92	8	8
Accountability	7.86	8	21
Storage limitation	7.17	7	30
Data minimisation	7.15	7	26
Accuracy	6.63	7	43

Sites score highest on principles **visible in the document itself** is there a clear policy, is a legal basis stated, is a contact provided. They score lowest on principles describing **restraint in practice**: collecting only what is needed, deleting it on time, keeping it correct.

Nearly half the corpus (43 of 89) scores below 7 on accuracy. This points us to the conclusion that the corpus is good at appearing compliant and weaker at being restrained. That is what one would expect if policies were written to satisfy a reader rather than to constrain an operator.

Figure 2: Sites score well on what you see





Discussion

For supervisory authorities, the absence of a machine-readable consent layer on 86% of sites (§4.4) is a structural obstacle to enforcement: auditing consent at scale requires that consent be expressible in a standard form. Separately, six of Europe's most-visited retailers cannot be inspected by standard automated means (§3.1).

For site operators, several major services in this corpus load zero third parties (§4.7). The prevailing pattern is not a technical requirement. The highest-leverage single change available to most operators is a first-layer reject control, currently absent on 71% of sites.

From the research side, the null result in §5.1 invites replication with a more discriminating scoring instrument, established inter-rater reliability, and a broader corpus.

And lastly, for people using the web. The finding here is not that you should read privacy policies more carefully. It is that reading them would not tell you what you want to know – and would cost you 25 minutes per site to discover that. The disclosure regime as currently practised does not equip an individual to make an informed choice.



Conclusion

The premise of European data-protection law is that disclosure enables choice: that a person, informed by the privacy policy and asked through the consent banner, can meaningfully decide.

This study tested that premise against evidence. The disclosure is unreadable, the user would need to spend 35 hours to read the privacy policies once and the consent is largely taken before it is asked, with no way to refuse on most sites. But the central finding is that the quality of what a company writes bears no measurable relationship to what its site does. Across three independent behaviours, the correlation is statistically indistinguishable from noise.

The results from this report are not evidence that policies lie, nor a claim of causation, and it is constrained by an instrument whose narrow scoring range attenuates the very correlation we sought. We show however that a policy quality cannot be used as a proxy for conduct. The document and the behaviour carry independent information, and only one of them can be measured without reading 35 hours of prose.

The compliance apparatus European users meet every day, such as the policy, the banner, the disclosure is not connected to the conduct it purports to govern.

Improving the disclosure by making policies clearer or banners friendlier, addresses the half of the problem that is already visible while leaving the half that matters untouched. If the goal is for people to know what is done with their data, the answer is not better documents. It is tools that allow them to own their decisions and understand their choices.

A well-written privacy policy is not evidence of privacy-respecting behaviour. It is evidence of a well-written privacy policy.



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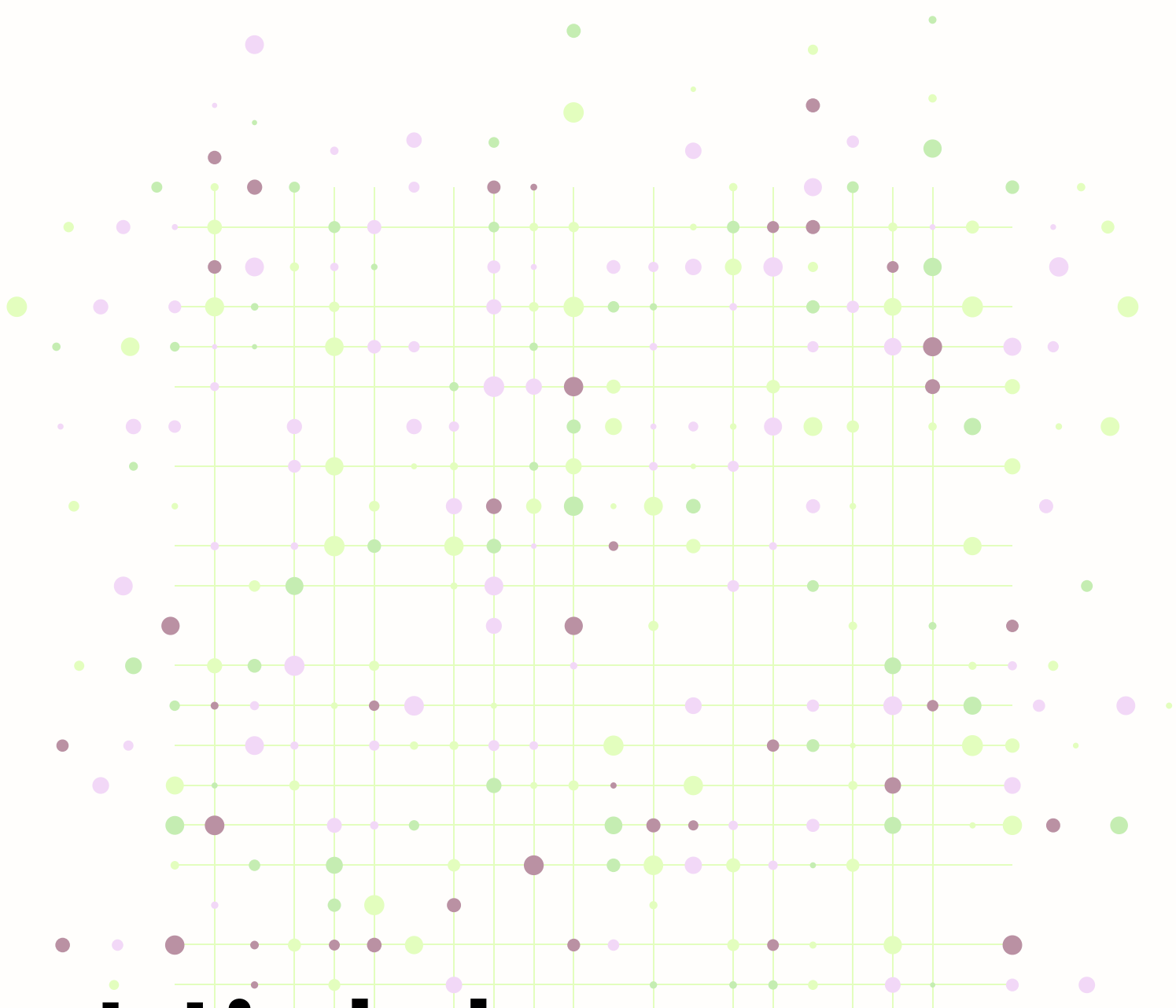
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